

PES OF CHEMICAL CHANGE

Name: _____

Date: _____

PROCEDURE:

When making observations be as descriptive as possible. These observations will assist you in the write up portion of your lab

STATION A

PART ONE

Place a small crystal of Copper (II) Sulfate Pentahydrate in a Pyrex test tube. Heat the test tube gently at a 45° angle using a Bunsen burner and a test tube holder. Make at least three distinctive qualitative observations during the heating process. **Safety: when heating point the mouth of the test tube away from classmates. Do not touch the test tube with your hands as it will get hot!**

PART TWO

Let the test tube and its contents sit until cool to the touch. You may place the test tube in a test tube rack for this part of the experiment. Using the water bottle provided add a few drops of water to the test tube and make your second set of observations for this station. **Safety: Do NOT add cold water to a hot test tube as you run the risk of shattering the glassware and splattering of contents.**

*** Pour the contents of the test tube into the waster beaker located in fume hood.***
Do NOT pour contents down the sink.

STATION B

Using sandpaper/steel wool scratch a 2 cm piece of copper. Using tongs heat the copper wire unit red for about 30 seconds in the hottest part of the flame (use a Bunsen Burner). After the wire has cooled scrape the surface of the wire with a scoopula.

*** CLEAN UP: PLACE THE PIECE OF COPPER IN THE GARBAGE CAN***

STATION C

PART ONE

In a test tube place TWO pieces of mossy zinc into a solution of 1 molar Hydrochloric acid (fill $\frac{3}{4}$ of the test tube with HCl).

PART TWO

Place the test tube in the test tube rack. Cover the mouth of the test tube with your thumb. When enough pressure has built up, so that you feel like your thumb is going to be pushed off, have your partner light a wooden splint and place it at the mouth of the test tube ('pop test'). Do not place the splint down the mouth of the test tube. **DO NOT WASTE SPLINTS.**

CLEAN UP: MOSSY ZINC FOR PART THREE.

PART THREE

Rub the wet mossy zinc against a piece of white paper.

***Clean up: Dry mossy zinc and place in beaker.

The mossy zinc can be re-used do NOT it through out!

STATION D:

Place some copper (II) sulfate in a test tube (half full) and then place an iron nail in the solution for about 2 minutes. Shake periodically and make observations.

Clean up: Rinse and dry the nail and place it back into its original container. Pour the copper (II) sulfate solution into the waster beaker located in the fume hood.

COLLECT WET

STATION E:

***** Do NOT contaminate pipettes*****

Part One

Using a pipette add a ONE milliliter of each solution, sodium sulfate and barium chloride, into a test tube. Make observations.

Part Two

Using a pipette add a ONE milliliter of each solution, potassium iodide and lead (II) nitrate, into a test tube. Make observations.

Part Three

Using a pipette add a ONE milliliter of each solution, sodium carbonate and calcium chloride, into a test tube. Make observations.

Clean up: Pour the solutions into the waste beaker located in the fume hood. NOT down the sink. Rinse and dry test tubes.

STATION F: (DEMO-TEACHER)

Fill a test tube half way with water and then place in test tube rack. Add 3-4 drops of indicator (phenolphthalein) note colour of solution in your observations. Using scupula place 1 to 2 pieces of Calcium (note colour of solution) and perform pop test.

STATION G: (DEMO-TEACHER)

Fill a test tube with Ethyne gas (C_2H_2) –teacher will demonstrate how this is achieved. Perform pop test.

Types of chemical change

Name: _____

Date: _____

(Application -10 marks) (title)

Station	Appearance of compound(s) (including state) before the experiment.	Observations
		1)
Station A (Part One)		2)
		3)
Station A (Part Two)		
Station B		
Station C (Part One)		
Station C (Part Two)		
Station C (Part Three)		
Station D		
Station E (Part One)		
Station E (Part Two)		
Station E (Part Three)		
Station F		
Station G		

Types of chemical change

Name: _____

Date: _____

POST LAB

Knowledge (11 marks)

1) For each station **name** the type of reaction that took place.

Station A :

Part One: _____ (1 mark)

Part Two: _____ (1 mark)

Station B :

Part One: _____ (1 mark)

Station C :

Part One: _____ (1 mark)

Part Two: _____ (1 mark)

Station D :

Part One: _____ (1 mark)

Station E :

Part One: _____ (1 mark)

Part Two: _____ (1 mark)

Part Three: _____ (1 mark)

Station F :

Part One: _____ (1 mark)

Station G :

Part One: _____ (1 mark)

Types of chemical change

Name: _____

Date: _____

TIPS (22 marks)

II) Write the chemical reaction(s) that took place at each station. Include all appropriate notation such as (s, l, g, aq and Δ).

Station A :

Part One: _____ (2 marks)

Part Two: _____ (2 marks)

Station B :

Part One: _____ (2 marks)

Station C :

Part One: _____ (2 marks)

Part Two: _____ (2 marks)

Station D :

Part One: _____ (2 marks)

Station E :

Part One: _____ (2 marks)

Part Two: _____ (2 marks)

Part Three: _____ (2 marks)

Station F :

Part One: _____ (2 marks)

Station G :

Part One: _____ (2 marks)